

Food Freshness Detection Using Machine Learning Algorithms

Dogga Pavan Sekhar

Department of Computer Science

Manav Rachna University

Faridabad, India

doggapavansekhar@gmail.com

Abstract—Maintaining food freshness is essential for reducing waste and ensuring food safety in modern households. This project presents a smart refrigerator system capable of automatically detecting and classifying the freshness of stored food items using image analysis and sensor data. The system integrates IoT-based temperature and humidity sensors with deep learning models to assess food conditions in real time. A labeled dataset of fruits and vegetables was used to train several state-of-the-art convolutional neural network (CNN) architectures, including VGG16, VGG19, ResNet50, DenseNet121, MobileNetV2, InceptionV3, Xception, and AlexNet. In addition to supervised learning, unsupervised clustering techniques such as K-Means and Agglomerative Clustering were employed to identify natural groupings within the data and enhance classification granularity. The freshness status is displayed on the smart refrigerator interface, enabling users to make timely decisions and reduce spoilage. This system demonstrates the potential of combining deep learning with IoT technologies to build intelligent, sustainable food management solutions.

Index Terms—Food Freshness Classification, Smart Refrigerator, CNN, K-Means Clustering, Agglomerative Clustering, Deep Learning, IoT Sensors.

I. INTRODUCTION

Food spoilage is a significant issue in both households and businesses, contributing to unnecessary food waste and health risks from consuming spoiled food. While refrigerators are commonly used to preserve food, they typically lack intelligent systems capable of assessing the condition of the food inside. As a result, users often rely on manual inspection, which can be inaccurate and delayed. With the rise of smart home technologies, integrating artificial intelligence (AI) into household appliances offers a promising solution to automate food monitoring and reduce waste.

This research, titled Food Freshness Detection, proposes a smart refrigerator system that uses image-based classification to monitor food freshness in real-time. A custom dataset was created, consisting of 13 common food categories, with each item labeled as either fresh or rotten. To assess freshness more accurately, clustering algorithms such as K-Means and Agglomerative Clustering were employed to categorize the food into five freshness levels: fresh, stale, aged, rotten, and highly spoiled. Agglomerative Clustering was found to produce the most coherent and interpretable groupings.

For the classification task, deep learning models, including MobileNet and ResNet, etc were used to classify food images into different freshness categories. These models analyze food images to determine freshness based on visual features. A system was developed to assess food freshness based on these images, allowing users to evaluate the condition of food stored in the refrigerator.

This research lays the foundation for intelligent food monitoring systems in smart refrigerators, offering a practical approach to food preservation. Future improvements may involve advanced detection techniques or additional methods to further enhance the system's overall effectiveness.

II. LITERATURE REVIEW

Food freshness detection is a vital area in food safety and waste reduction. Traditional methods such as sensory evaluation, microbial testing, and expiration dates are often imprecise and lead to either unnecessary waste or health risks. In [1], the Fresh Sight system introduces a user-friendly AI-powered approach that supports both standard and color-blind users. It uses visual cues and deep learning algorithms to assess produce freshness based on color features and texture. The model achieved high accuracy in classifying freshness levels and proved practical for real-time use. However, the authors noted limitations like dependency on lighting conditions and potential bias in datasets collected under controlled environments.

Machine learning (ML) models have shown strong potential in predicting food spoilage more accurately. In [3], Shehzad presents a comprehensive overview of predictive AI systems that incorporate various data sources such as temperature, humidity, gas emissions, and microbial activity. The paper highlights the use of Convolutional Neural Networks (CNNs) and hybrid models combining machine learning and deep learning for classifying food as fresh or spoiled. These systems outperform traditional estimation techniques and allow dynamic shelf-life prediction. Nevertheless, challenges remain in generalizing models across food types due to data variability and infrastructure limitations in smaller industries.

The role of sensors and smart storage systems is emphasized in [2], where a Resnet-based model is used for refrigerator optimization. The study focused on enhancing internal storage

efficiency by predicting food categories and temperature needs. Although not directly assessing spoilage, this model supports freshness indirectly by improving food placement and storage dynamics. The system, while effective, requires consistent sensor performance and integration with AI modules to provide holistic freshness assessment.

Advanced sensor technologies combined with deep learning were explored in [4], where the authors deployed a fluorescent sensing array to monitor meat freshness in real-time. Their deep learning model showed high responsiveness and accuracy, making it suitable for industrial deployment. The model was trained to recognize chemical changes indicating spoilage, offering a non-invasive and fast method. While promising, such setups require specialized equipment and may not be easily scalable to all food types or retail settings.

Finally, biosensing technologies enhanced with machine learning are discussed in [5], where Zhou et al. demonstrate how biosensors can detect spoilage indicators at the molecular level. Machine learning enhances the interpretation of biosensor data, improving detection speed and accuracy. These methods could revolutionize food safety, especially when integrated into packaging systems. However, affordability and complexity of biosensor manufacturing remain key barriers to large-scale adoption. These studies collectively show that AI, deep learning, and smart sensors significantly improve food freshness detection. While some models prioritize visual analysis [1], others leverage environmental and chemical data [3–5] for more comprehensive assessment. Still, consistent data quality, scalability, and integration challenges must be addressed before these technologies become mainstream in households and supply chains.

III. DATASET DESCRIPTION

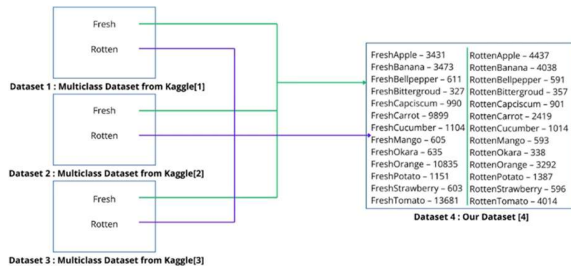


Fig. 1. Mapping of source datasets (Dataset 1, Dataset 2, Dataset 3) to the final combined dataset used for food freshness detection. This figure visualizes the integration process and demonstrates how the datasets contribute to the overall balanced structure of the dataset.

A combined food image dataset was developed using three publicly available sources (Datasets [1], [2], and [3]). The images were organized into two primary classes Fresh and Rotten each comprising 13 identical subcategories of common fruits and vegetables (e.g., apple, banana, tomato), resulting in 26 subfolders. To maintain class balance, 100 images were selected per subfolder, producing a total of 2,600 images. Labels were assigned based on the dataset of origin: Label 1,

Label 2, and Label 3 represent the respective original dataset sources, with each assigned based on food type and freshness level. Label 4 represents the integrated dataset, maintaining a balanced distribution of food images.

IV. PROPOSED METHODOLOGY

A. Methodology

The proposed system integrates IoT sensors and deep learning techniques to automate food freshness detection within a smart refrigerator. As depicted in Figure 2, the system consists of IoT sensors that monitor temperature and humidity, a camera to capture high-resolution images of the food, a processing unit for data analysis, and a display interface for user interaction. The captured data undergoes preprocessing before being fed into a trained deep learning model, which classifies food freshness. The results are then displayed on the interface, enabling users to make informed decisions that reduce food waste and optimize consumption.

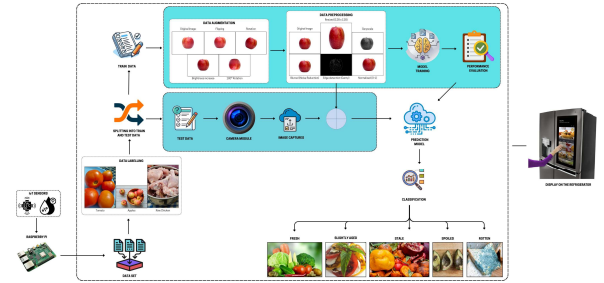


Fig. 2. Illustrates the overall system components and how they interact in the context of the proposed methodology.

The following sections provide a detailed explanation of how each block in the diagram aligns with the corresponding steps outlined in the methodology.

The data acquisition process captures high-resolution food images under varying conditions, with IoT sensors monitoring temperature and humidity. Images are labeled into five freshness levels—Very Fresh, Fresh, Slightly Aged, Stale, and Rotten—based on visual cues, with expert validation ensuring accuracy. The images are preprocessed through resizing, normalization, and denoising, followed by data augmentation techniques to improve model generalization. The dataset is split into 80

B. Clustering Methods

Unsupervised clustering algorithms were used to classify food items based on their visual freshness attributes, minimizing the need for manual labeling. The following algorithm was implemented:

1) **K-Means Clustering:** K-Means clustering was applied to categorize food images into freshness levels, such as Very Fresh, Fresh, Slightly Aged, Stale, and Rotten. It works well when the number of clusters is predetermined, offering a simple and computationally efficient solution for large datasets. However, K-Means is sensitive to the initial choice of centroids

and may struggle with non-spherical clusters or varying cluster densities.

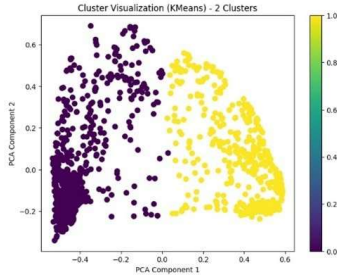


Fig. 3. 2 Clusters

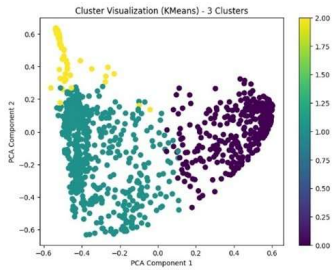


Fig. 4. 3 Clusters

Figures 3 and 4 illustrate how K-Means clustering with 2 and 3 clusters visually separates food items based on freshness. These clusters align with freshness categories, providing an unsupervised way to categorize food images before supervised classification.

2) **Agglomerative Clustering** : Agglomerative clustering, in contrast, builds clusters hierarchically, which provides greater flexibility in terms of cluster shape and size. It is especially useful when the number of clusters is not known in advance. However, this method can become computationally expensive as the dataset grows, making it less scalable compared to K-Means. Figures 5 and 6 present Agglomerative Clustering

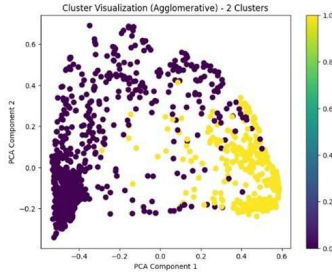


Fig. 5. 2 Clusters

results with 2 and 3 clusters using PCA for visualization. The clear separation highlights how clustering structure varies with the number of clusters.

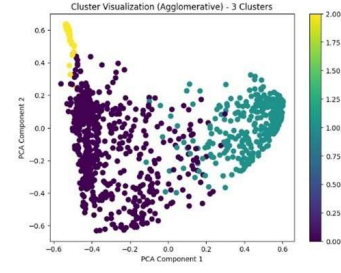


Fig. 6. 3 Clusters

C. Proposed Models

Various advanced deep learning architectures have been explored for accurate food freshness classification, each offering unique strengths in feature extraction, learning capacity, and computational efficiency. This comparative analysis aims to identify the model that balances predictive performance with real-time deployment feasibility in a smart refrigerator system. Both lightweight and deeper models were considered to address the complexities of food spoilage detection.

The following sections provide an overview of the deep learning models evaluated in this study. Each model's architecture, strengths, and suitability for food freshness detection are discussed, highlighting their performance in capturing relevant features and their feasibility for integration into a smart refrigerator system.

1) **VGG16**: It is a deep CNN with 16 layers using 3x3 convolution filters. It excels at capturing fine-grained features, making it effective for food freshness detection, especially in texture and color variations. Despite its computational demands, VGG16 is ideal for distinguishing subtle differences in freshness.

2) **VGG19**: It extends VGG16 by adding more layers, enabling deeper feature extraction. It is used in this project to detect complex spoilage patterns that require increased depth. VGG19's higher accuracy justifies its computational cost for advanced food freshness classification tasks.

3) **MobileNetV2** : It is a lightweight CNN optimized for efficiency, using depth wise separable convolutions. It is ideal for real-time applications in smart refrigerators, balancing feature extraction and efficiency for low-latency, low-power food freshness detection in embedded systems.

4) **ResNet50**: It is a deep CNN with 50 layers and residual connections, which address the vanishing gradient problem. It is used for its ability to learn hierarchical features and identify subtle spoilage patterns, ensuring high accuracy in food freshness detection, despite its computational intensity.

5) **DenseNet121**: The features densely connected layers that enhance feature reuse and gradient flow. It is employed for its efficient learning of complex spoilage patterns, offering a good balance of accuracy and computational efficiency for detailed food freshness classification.

6) **InceptionV3**: It uses multi-scale processing with parallel convolutional layers, enabling it to extract features at varying

spatial scales. It is used for its ability to detect both local and global spoilage features in food images, ensuring robust freshness classification across varying spoilage levels.

7) **Xception**: It replaces traditional convolutions with depth wise separable convolutions, improving efficiency and performance. It is used for its high accuracy in capturing complex spoilage patterns, making it suitable for diverse food types and fine-tuning in freshness detection tasks.

8) **AlexNet**: It is a deep CNN with five convolutional layers, suitable for basic image classification. It serves as a baseline model in this project, enabling quick evaluation of food freshness detection performance before advancing to more complex models for improved accuracy.

V. RESULTS AND DISCUSSIONS

A. Comparison of Results

TABLE I
PERFORMANCE COMPARISON OF PRETRAINED MODELS WITH K-MEANS CLUSTERING

Model	Accuracy	Macro Precision	Macro Recall	Macro F1-Score
AlexNet	0.72	0.71	0.72	0.70
DenseNet121	0.75	0.73	0.74	0.74
InceptionV3	0.74	0.74	0.72	0.72
MobileNetV2	0.77	0.75	0.76	0.75
ResNet50	0.37	0.20	0.32	0.23
VGG16	0.66	0.65	0.63	0.64
VGG19	0.68	0.66	0.65	0.65
Xception	0.72	0.70	0.70	0.70

TABLE II
PERFORMANCE COMPARISON OF PRETRAINED MODELS WITH AGGLOMERATIVE CLUSTERING

Model	Accuracy	Macro Precision	Macro Recall	Macro F1-Score
AlexNet	0.84	0.82	0.86	0.83
DenseNet121	0.83	0.81	0.83	0.82
InceptionV3	0.82	0.82	0.83	0.83
MobileNetV2	0.81	0.80	0.81	0.80
ResNet50	0.74	0.73	0.73	0.73
VGG16	0.70	0.56	0.59	0.57
VGG19	0.66	0.53	0.55	0.53
Xception	0.42	0.34	0.34	0.34

Table 1 presents the performance of various pretrained CNN models integrated with K-Means clustering for food freshness detection. Among all models, MobileNetV2 achieved the highest performance with an accuracy of 0.77 and a macro F1-score of 0.75, followed closely by DenseNet121 and InceptionV3. ResNet50 performed poorly across all metrics, indicating its unsuitability for this task. Models like AlexNet, Xception, and VGG variants showed moderate results, with accuracies ranging from 0.66 to 0.72. This comparison highlights MobileNetV2 as the most effective model for the given clustering-based classification problem.

Table 2 displays the evaluation results of various pretrained models combined with Agglomerative Clustering for food freshness detection. AlexNet achieved the highest performance with an accuracy of 0.84 and macro F1-score of 0.83, followed closely by InceptionV3 and DenseNet121, both showing

strong and consistent metrics. MobileNetV2 also performed well with an F1-score of 0.80. In contrast, Xception and VGG19 recorded the lowest scores, indicating poor clustering compatibility. The results suggest that AlexNet, DenseNet121, and InceptionV3 are most effective for this task under Agglomerative Clustering.

VI. CONCLUSION AND FUTURE WORK

This project presents a smart food freshness detection system that integrates IoT sensors with deep learning models to classify food items based on their freshness levels. By leveraging image data and environmental sensor readings, the system achieves accurate classification of fruits and vegetables into subcategories such as fresh, very fresh, aged, stale, and spoiled. Among the models tested, MobileNetV2 and AlexNet demonstrated high performance in both accuracy and efficiency, making them suitable for real-time applications in smart refrigeration.

For future enhancements, the system can be extended by incorporating additional sensor types, such as gas sensors, to improve spoilage detection. Implementation on embedded hardware like Raspberry Pi, along with mobile app integration for user-friendly alerts and monitoring, is also planned. Moreover, expanding the dataset to include more food types and environmental conditions will further improve the model's robustness and scalability in practical use.

REFERENCES

- [1] Fernando, P., Kirupananda, A. FreshSight: A colour-blind-friendly solution for assessing produce freshness, ensuring food safety, and reducing waste.
- [2] Khaleel, Yahya Layth, Mustafa Abdulfattah Habeeb, and M. A. Ahmed. "Refrigerator optimization: Leveraging RESnet method for enhanced storage efficiency." In AIP Conference Proceedings, vol. 3264, no. 1, p. 040009. AIP Publishing LLC, 2025.
- [3] Shehzad, K., 2025. Predictive AI Models for Food Spoilage and Shelf-Life Estimation. Global Trends in Science and Technology, 1(1), pp.75-94.
- [4] Lin, Y., Ma, J., Sun, D.W., Cheng, J.H. and Zhou, C., 2024. Fast real-time monitoring of meat freshness based on fluorescent sensing array and deep learning: From development to deployment. Food chemistry, 448, p.139078.
- [5] Mogilipalem, Keerthana, et al. "Food spoilage detection using IoT and machine learning." 2024 5th International Conference for Emerging Technology (INCET). IEEE, 2024.
- [6] Yuan, Yue, and Xianlong Chen. "Vegetable and fruit freshness detection based on deep features and principal component analysis." Current Research in Food Science 8 (2024): 100656.
- [7] Vasanthi, R., K. Ashika, and D. Archana. "IoT-based Food Quality Monitoring System Using Machine Learning." 2024 2nd International Conference on Networking and Communications (ICNWC). IEEE, 2024.
- [8] Zhang, Yinsheng, et al. "Fruit freshness detection based on multi-task convolutional neural network." Current Research in Food Science 8 (2024): 100733.
- [9] Abayomi-Alli, Olusola O., et al. "FruitQ: a new dataset of multiple fruit images for freshness evaluation." Multimedia Tools and Applications 83.4 (2024): 11433-11460.
- [10] Bhuiyan, Zarif Wasif, et al. "IoT Based Meat Freshness Classification Using Deep Learning." IEEE Access (2024).
- [11] Vasanthi, R., K. Ashika, and D. Archana. "IoT-based Food Quality Monitoring System Using Machine Learning." 2024 2nd International Conference on Networking and Communications (ICNWC). IEEE, 2024.
- [12] Han, Junming, et al. "Using machine learning approaches for food quality detection." Mathematical Problems in Engineering 2022.1 (2022): 6852022.

- [13] Anwar, Hassan, Talha Anwar, and Shamas Murtaza. "Review on food quality assessment using machine learning and electronic nose system." *Biosensors and Bioelectronics*: X 14 (2023): 100365.
- [14] Singh, Raj, et al. "A Comprehensive Review of Advanced Deep Learning Approaches for Food Freshness Detection." *Food Engineering Reviews* (2024): 1-34.
- [15] Hindarto, Djarot. "Model Performance Evaluation: VGG19 and Dense201 for Fresh Meat Detection." *Sinkron: jurnal dan penelitian teknik informatika* 8.1 (2024): 514-524.
- [16] <https://www.kaggle.com/datasets/alinesellwia/food-freshness>
- [17] <https://www.kaggle.com/datasets/swoyam2609/fresh-and-stale-classification>
- [18] <https://www.kaggle.com/datasets/muhriddinmuxiddinov/fruits-and-vegetables-dataset>
- [19] <https://www.kaggle.com/datasets/ulnnproject/food-freshness-dataset>
- [20] Mallegowda, M., et al. "Fruit classification based on freshness." 2024 International Conference on Emerging Techniques in Computational Intelligence (ICETCI). IEEE, 2024.